

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical

Task

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Extra Task:

Perform following questions using Retail.csv

Q.1 Write a Python program to:

Count the number of products in each Product Category.

Code:-

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("Retail_sales.csv")
# Count products in each category
count = df["Product Category"].value_counts()
print("Products in Each Category:")
print(count)
```

Output:-

```
Products in Each Category:
Product Category
Furniture      9503
Electronics    8041
Clothing       6608
Groceries      5848
Name: count, dtype: int64
```

Find the category with the highest sales.

Code:-

```
# Sales by category
sales = df.groupby("Product Category")["Sales Revenue (USD)"].sum()
print("\nSales by Category:")
print(sales)
```

Output:-

```
Sales by Category:
Product Category
Clothing      19952024.73
Electronics   29414540.61
Furniture     22913105.64
Groceries     10205616.80
Name: Sales Revenue (USD), dtype: float64
```

Highest sales category

```
print("\nCategory with Highest Sales:")
print(sales.idxmax())
```

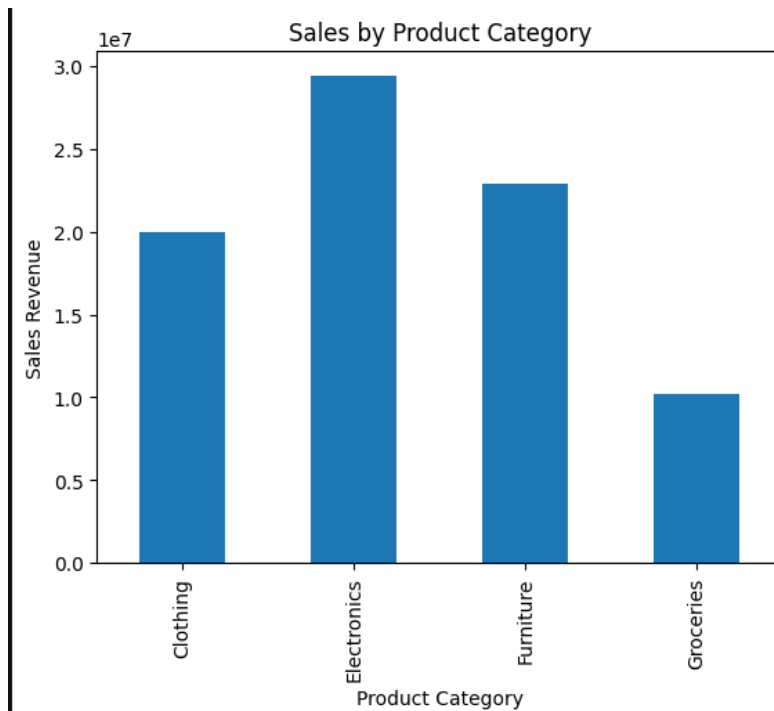
```
Category with Highest Sales:  
Electronics
```

Display the result using a bar chart.

Code:-

```
# Bar Chart  
sales.plot(kind="bar")  
plt.title("Sales by Product Category")  
plt.xlabel("Product Category")  
plt.ylabel("Sales Revenue")  
plt.show()
```

Output:-



Q.2 Write a Python program to:

Calculate the total Sales Revenue for each Store ID.

Find the best-performing store.

Find the least-performing store.

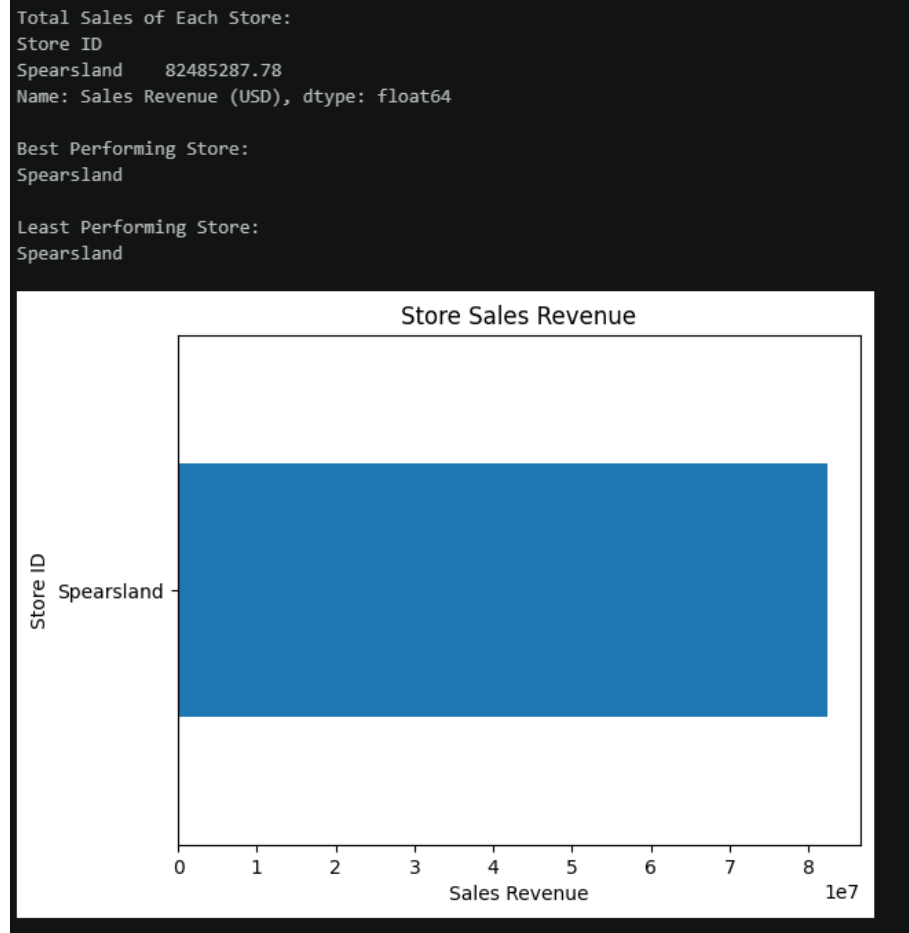
Display the result using a horizontal bar chart.

Code:-

```
store = df.groupby("Store ID")["Sales Revenue (USD)"].sum()  
  
print("Total Sales of Each Store:")  
print(store)  
  
print("\nBest Performing Store:")  
print(store.idxmax())  
  
print("\nLeast Performing Store:")  
print(store.idxmin())
```

```
# Horizontal Bar Chart
store.plot(kind="barh")
plt.title("Store Sales Revenue")
plt.xlabel("Sales Revenue")
plt.ylabel("Store ID")
plt.show()
```

Output:-



Q.3 Write a Python program to:

Calculate total Sales Revenue for each Store Location.

Identify the location with the highest sales.

Plot the results using a bar chart.

Code:-

```
location = df.groupby("Store Location")["Sales Revenue (USD)"].sum()
```

```
print(location)
```

```
print("\nLocation with Highest Sales:")
```

```
print(location.idxmax())
```

Bar Chart

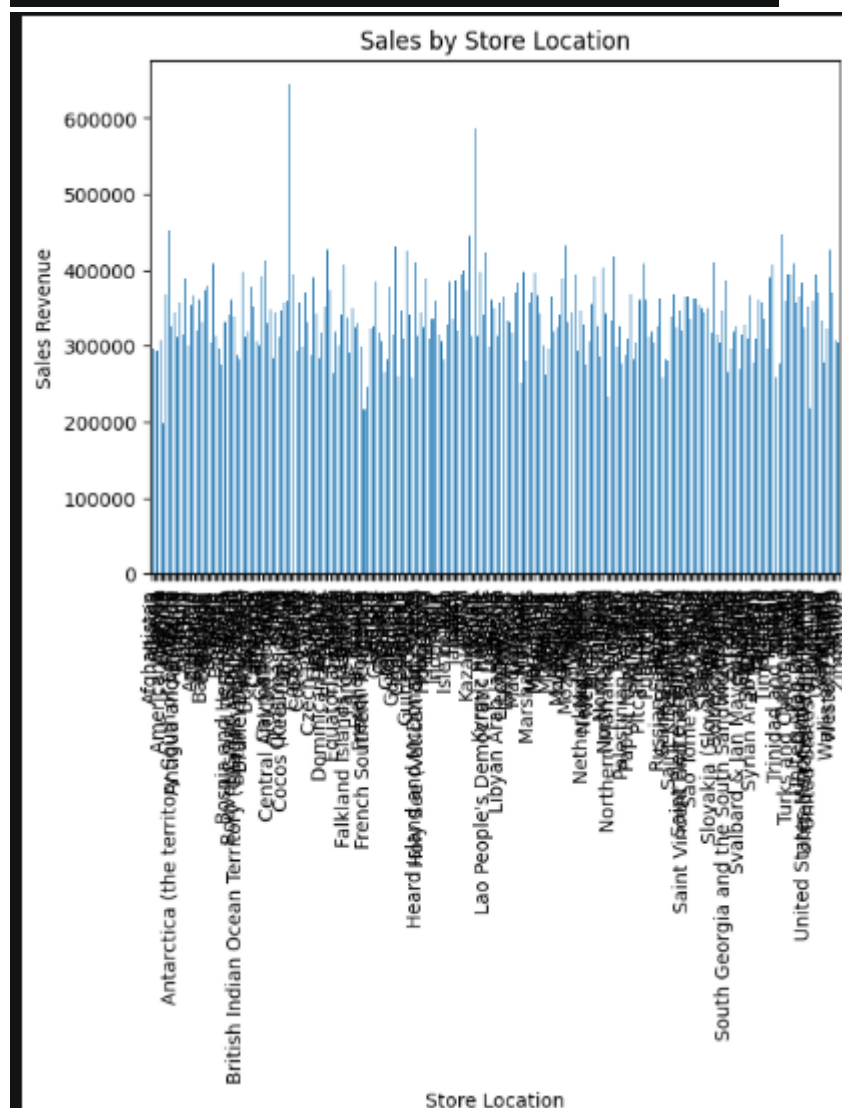
```
location.plot(kind="bar")
plt.title("Sales by Store Location")
plt.xlabel("Store Location")
plt.ylabel("Sales Revenue")
```

```
plt.show()
```

Output:-

```
Store Location
Afghanistan      329511.22
Albania          296929.47
Algeria          292690.46
American Samoa   308614.68
Andorra          199063.87
...
Wallis and Futuna 323061.14
Western Sahara   426248.78
Yemen           369518.90
Zambia          307427.30
Zimbabwe        305188.53
Name: Sales Revenue (USD), Length: 243, dtype: float64

Location with Highest Sales:
Congo
```



Q.4 Calculate the following descriptive statistics for the Units Sold column:

Mean

Median

Mode

Minimum

Maximum

Range

Variance

Standard Deviation

Code:-

```
x = df["Units Sold"]
```

```
print("Mean =", x.mean())
print("Median =", x.median())
print("Mode =", x.mode()[0])
print("Minimum =", x.min())
print("Maximum =", x.max())
print("Range =", x.max() - x.min())
print("Variance =", x.var())
print("Standard Deviation =", x.std())
```

Output:-

```
Mean = 6.161966666666666
Median = 6.0
Mode = 7
Minimum = 0
Maximum = 56
Range = 56
Variance = 11.048501748947187
Standard Deviation = 3.323928661831837
```

Perform following operations on Netflix.csv

Q.1 Create a Box Plot for the Release Year column.

Identify:

Minimum value

Maximum value

Median

Presence of outliers

Code:-

```
# Read Dataset
df = pd.read_csv("NetFlix.csv")
plt.boxplot(df["release_year"])
plt.title("Box Plot of Release Year")
plt.show()

print("Minimum =", df["release_year"].min())
print("Maximum =", df["release_year"].max())
print("Median =", df["release_year"].median())

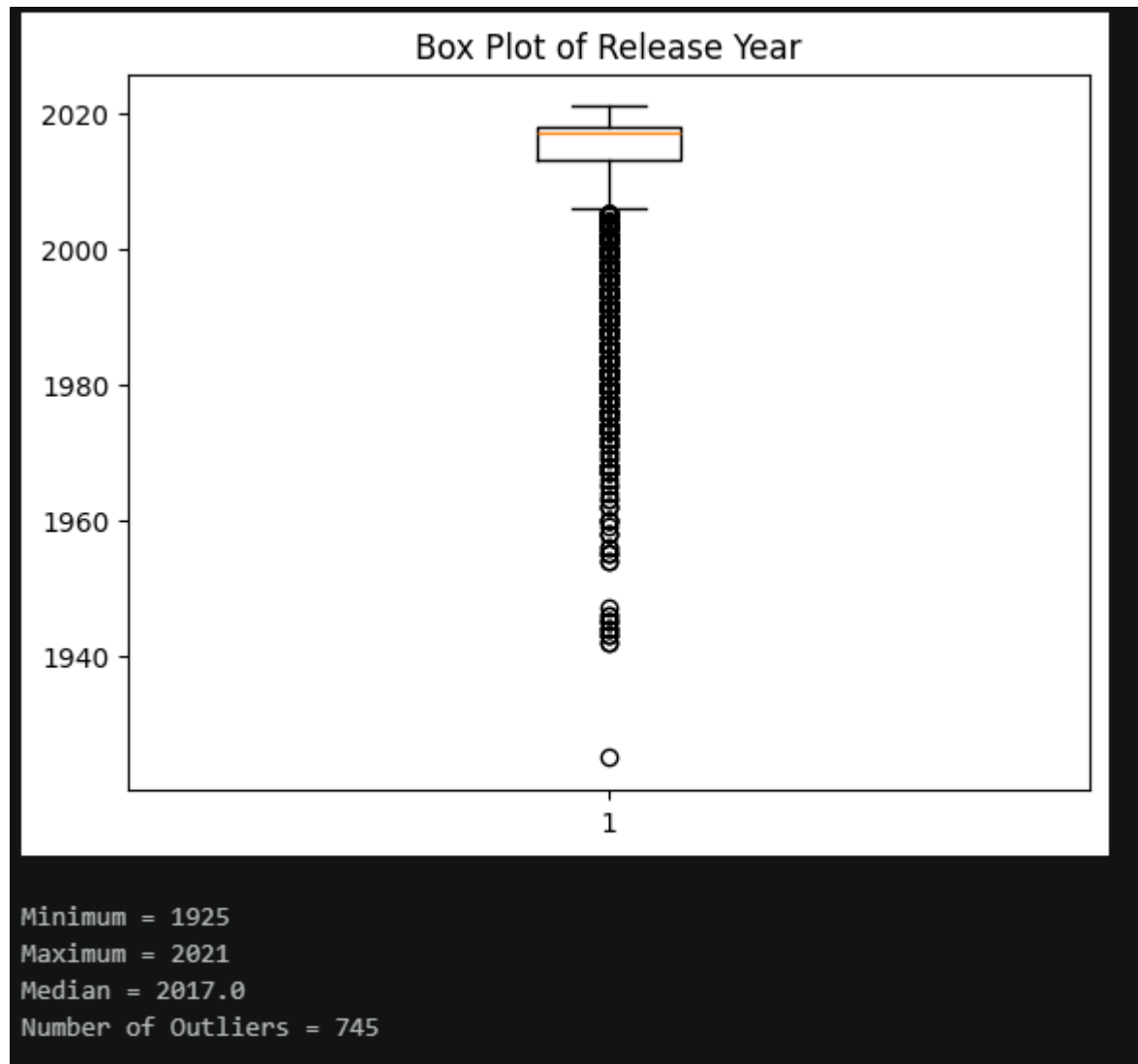
Q1 = df["release_year"].quantile(0.25)
Q3 = df["release_year"].quantile(0.75)
IQR = Q3 - Q1
```

```
lower = Q1 - 1.5 * IQR  
upper = Q3 + 1.5 * IQR
```

```
outliers = df[(df["release_year"] < lower) | (df["release_year"] > upper)]
```

```
print("Number of Outliers =", len(outliers))
```

Output:-



Q.2 Filter only Movies from the dataset.

Extract the numerical value from the Duration column (e.g., 90 min → 90).

Calculate:

Mean

Median

Mode

Variance

Standard Deviation

for movie duration.

Code:-

```
# Keep only Movies  
movies = df[df["type"] == "Movie"].copy()
```

Convert duration column to string before extracting numbers

```
movies["duration"] = movies["duration"].astype(str).str.extract(r"(\d+)").astype(float)
```

Calculate statistics

```
print("Mean =", movies["duration"].mean())
```

```
print("Median =", movies["duration"].median())
```

```
print("Mode =", movies["duration"].mode()[0])
```

```
print("Variance =", movies["duration"].var())
```

```
print("Standard Deviation =", movies["duration"].std())
```

Output:-

```
Mean = 99.30797842663195
```

```
Median = 98.0
```

```
Mode = 90.0
```

```
Variance = 814.0111584310822
```

```
Standard Deviation = 28.53088078610757
```